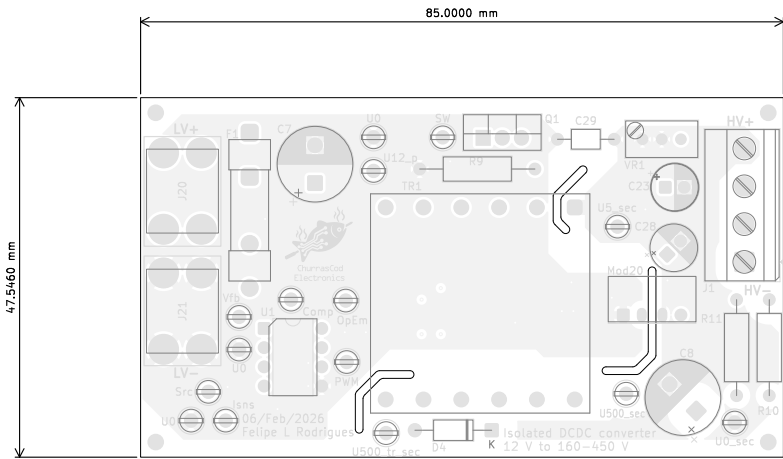


Module Fabrication Document

Top Fabrication (Scale 1:1)



All dimensions are in millimeters unless otherwise specified.

FABRICATION NOTES (UNLESS OTHERWISE SPECIFIED)

- 1) FABRICATE PER IPC-6012A CLASS 2.
- 2) OUTLINE DEFINED IN SEPARATE GERBER FILE WITH "Edge_Cuts.GBR" SUFFIX.

DIMENSIONS OF CIRCUMSIZED RECTANGLE SHOWN ON THIS DRAWING ARE FOR REFERENCE ONLY.
- 3) SEE SEPARATE DRILL FILES WITH ".DRL" SUFFIX FOR HOLE LOCATIONS.

SELECTED HOLE LOCATIONS SHOWN ON THIS DRAWING ARE FOR REFERENCE ONLY.
- 4) SURFACE FINISH: IMMERSION GOLD
- 5) SOLDERMASK ON BOTH SIDES OF THE BOARD SHALL BE LPI, COLOR TOP: #8071501A / BOTTOM: GREEN.
- 6) SILK SCREEN LEGEND TO BE APPLIED PER LAYER STACKUP USING WHITE NON-CONDUCTIVE EPOXY INK.
- 7) ALL VIAS ARE TENTED ON BOTH SIDES UNLESS SOLDERMASK OPENED IN GERBER.
- 8) VENDOR SHOULD FOLLOW ROHS COMPLIANT PROCESS AND Pb FREE FOR MANUFACTURING
- 9) PCB MATERIAL REQUIREMENTS:

A. FLAMMABILITY RATING MUST MEET OR EXCEED UL94V-0 REQUIREMENTS.
B. Tg 170 C OR EQUIVALENT.
C. EQUIVALENT MATERIAL SHALL BE RoHS COMPLIANT, HALOGEN FREE AND APPROVED BY FR.

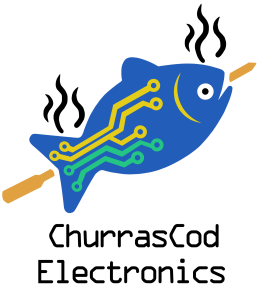
10) DESIGN GEOMETRY MINIMUM FEATURE SIZES:

BOARD SIZE	85,000 × 47,546 mm
BOARD THICKNESS	1.660 mm
TRACE WIDTH	0.400 mm
TRACE TO TRACE	-0,000 mm
MIN. HOLE (PTH)	0.250 mm
MIN. HOLE (NPTH)	2.100 mm
ANNULAR RING	0.150 mm
COPPER TO HOLE	0.200 mm
COPPER TO EDGE	0.250 mm
HOLE TO HOLE	0.254 mm

Layer Stack Legend

Material	Layer	Thickness	Dielectric	Type	Gerber
F.Paste				Paste Mask	
F.Silkscreen				Legend	GBR
F.Mask		0.02mm	Solder Resist	Solder Mask	GBR
Copper	L1 (Ref)	0.07mm (2.00oz)		Plane	GBR
Prepreg		0.1mm	FR4_7628	Dielectric	
Copper	Virtual (Top slde jumpers)	0.035mm (1.00oz)		Copper Layer	GBR
Core		1.21mm	FR4	Dielectric	
Copper	Virtual (Bot slde jumpers)	0.035mm (1.00oz)		Copper Layer	GBR
Prepreg		0.1mm	FR4	Dielectric	
Copper	L2 (Sig, PWR)	0.07mm (2.00oz)		Signal	GBR
B.Mask		0.02mm	Solder Resist	Solder Mask	GBR
B.Silkscreen				Legend	GBR
B.Paste				Paste Mask	

Total thickness: 1.66mm
Note: external layer thicknesses are specified after plating



Comments:

- Comment 1
- Comment 2
- Comment 3
- Comment 4

Sheet Title:

Top Fabrication (Scale 1:1)

Sheet Path:

Company:

FR

Board Name:

Module

File Name:

Isolated_12to450V.kicad_pcb

Designer:

FR

Reviewer:

NA

Variant:

PRELIMINARY

Project Name:

Isolated step-up 12:450 V

Date:

2026-02-02

Size:

A4

Git Hash:

78dc7f5

Revision:

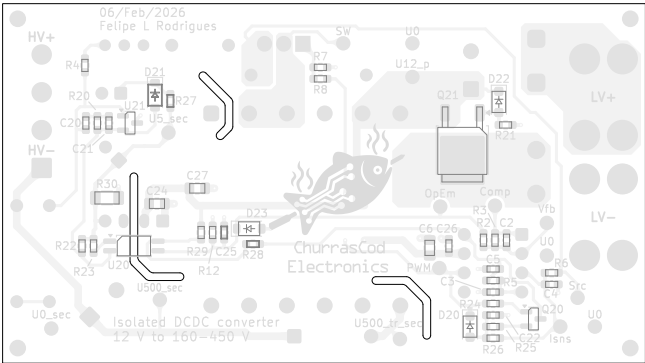
1.0.0+ (Unreleased)

Revision:

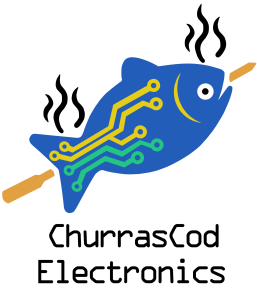
1.0.0+ (Unreleased)

Sheet:

1 of 7



All dimensions are in millimeters unless otherwise specified.



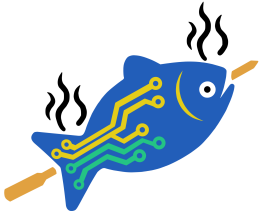
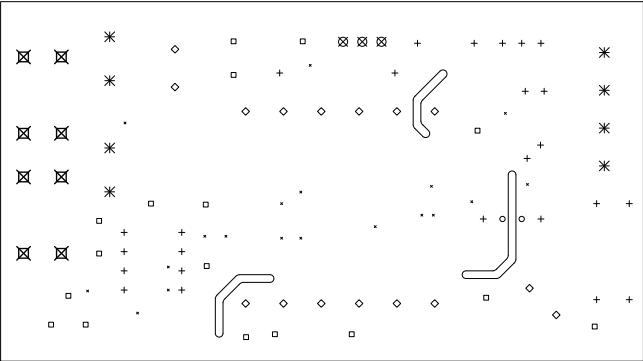
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Comment 1	FR		PRELIMINARY	78dc7f5
Comment 2	Board Name: Module		Project Name:	
Comment 3			Isolated step-up 12:450 V	
Comment 4				
Sheet Title:	File Name:	Designer:	Date:	Revision:
Bottom Fabrication (Scale 1:1)	Isolated_12to450V.kicad_pcb	FR	2026-02-02	1.0.0+ (Unreleased)
Sheet Path:		Reviewer:	Size:	Sheet:
		NA	A4	2 of 7

Module Fabrication Document

Drill Drawing L1 - L4 (Scale 1:1)

Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	19	0,25mm (9,84mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Via
○	2	0,70mm (27,56mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Pad
+	25	0,80mm (31,50mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Pad
□	17	0,90mm (35,43mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Pad
◇	16	1,00mm (39,37mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Pad
⊗	3	1,20mm (47,24mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Pad
※	8	1,30mm (51,18mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Pad
⊠	8	1,80mm (70,87mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Pad
Total 98						



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Comments:

Comment 1
Comment 2
Comment 3
Comment 4

Sheet Title:

Drill Drawing (L1 - L4)

Sheet Path:

Company:

FR

Board Name:

Module

File Name:

Isolated_12to450V.kicad_pcb

Variant:

PRELIMINARY

Git Hash:

78dc7f5

Project Name:

Isolated step-up 12:450 V

Designer:

FR

Reviewer:

NA

Date:

2026-02-02

Size:

A4

Revision:

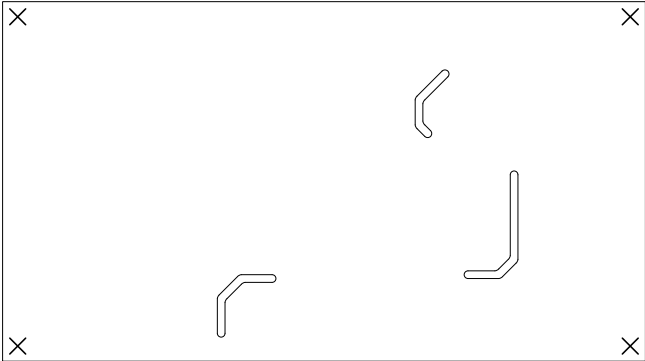
1.0.0+ (Unreleased)

Sheet:

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Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
X	4	2.10mm (82.68mil)	NPTH	Round	L1 (Ref) - L2 (Sig, PWR)	Mechanical
	Total 4					

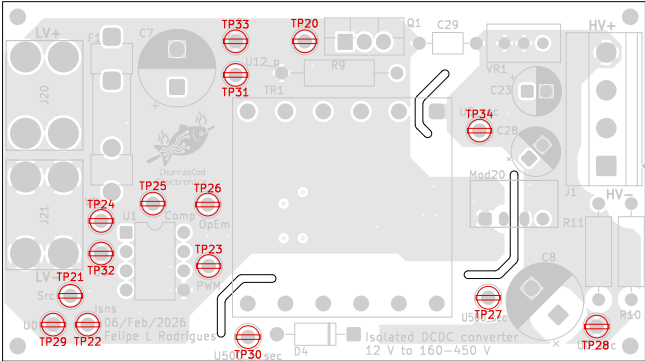


Comments: Comment 1 Comment 2 Comment 3 Comment 4	Company: FR		Variant: PRELIMINARY	Git Hash: 78dc7f5
	Board Name: Module		Project Name: Isolated step-up 12:450 V	
Sheet Title: Drill Drawing (L1 - L4)	File Name: Isolated_12to450V.kicad_pcb	Designer: FR	Date: 2026-02-02	Revision: 1.0.0+ (Unreleased)
Sheet Path:		Reviewer: NA	Size: A4	Sheet: 4 of 7

Module Fabrication Document

Top Test Points (Scale 1:1)

Ref.	Net	X [mm]	Y [mm]
TP20	SW	41.95	44.73
TP21	Src	10.97	11.07
TP22	Isns	13.25	7.26
TP23	PWM	29.25	15.01
TP24	Vfb	15.03	20.98
TP25	Comp	21.89	23.27
TP26	OpEm	29.13	23.14
TP27	U500_sec	66.21	10.82
TP28	U0_sec	80.56	7.01
TP29	U0	8.68	7.26
TP30	U500_tr_sec	34.46	5.61
TP31	U12_p	32.81	40.28
TP32	U0	15.03	16.66
TP33	U0	32.81	44.73
TP34	U5_sec	65.07	32.92



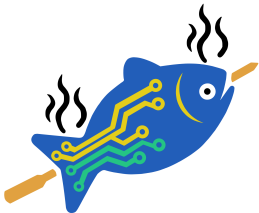
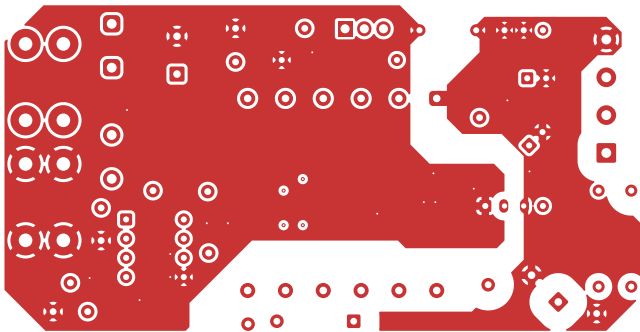
All dimensions are in millimeters unless otherwise specified.

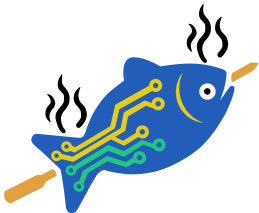


	Comments: Comment 1 Comment 2 Comment 3 Comment 4		Company: FR		Variant: PRELIMINARY	Git Hash: 78dc7f5
	Sheet Title: Top Test Points (Scale 1:1)		Board Name: Module		Project Name: Isolated step-up 12:450 V	
	Sheet Path:		File Name: Isolated_12to450V.kicad_pcb	Designer: FR	Date: 2026-02-02	Revision: 1.0.0+ (Unreleased)
			Reviewer: NA	Size: A4	Sheet: 5 of 7	

Module Fabrication Document

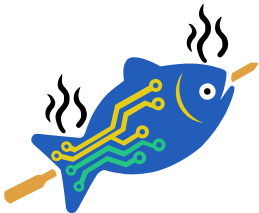
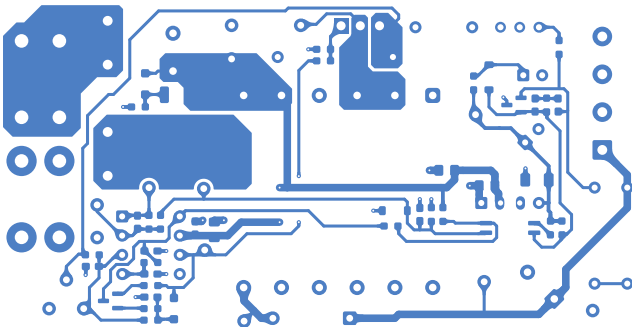
F.Cu (Scale 1:1)



	Comments: Comment 1 Comment 2 Comment 3 Comment 4		Company: FR		Variant: PRELIMINARY	Git Hash: 78dc7f5
	Sheet Title: F.Cu (Scale 1:1)		Board Name: Module		Project Name: Isolated step-up 12:450 V	
	Sheet Path:		File Name: Isolated_12to450V.kicad_pcb	Designer: FR	Date: 2026-02-02	Revision: 1.0.0+ (Unreleased)
			Reviewer: NA	Size: A4	Sheet: 6 of 7	

Module Fabrication Document

B.Cu (Scale 1:1)



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Comments:

- Comment 1
- Comment 2
- Comment 3
- Comment 4

Sheet Title:

B.Cu (Scale 1:1)

Sheet Path:

Company:

FR

Board Name:

Module

File Name:

Isolated_12to450V.kicad_pcb

Designer:

FR

Reviewer:

NA

Variant:

PRELIMINARY

Git Hash:

78dc7f5

Project Name:

Isolated step-up 12:450 V

Date:

2026-02-02

Revision:

1.0.0+ (Unreleased)

Size:

A4

Sheet:

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